

Lecture 16

$$P = A(A^T A)^{-1} A^T$$

- If b is in the column space
 $Pb = b$ [Q. how when A is not sq. inv.]

- If b is $\perp$ to the column space.

$$Pb = \{0\}$$

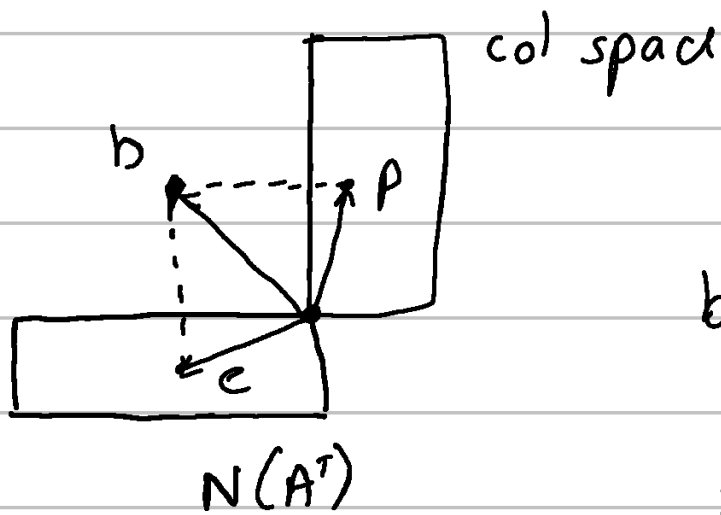
$$Pb = A(A^T A)^{-1} \underbrace{A^T b}_{= 0} = 0$$

Proof for the first part $\rightarrow$

$$b = Ax$$

$$Pb = A \underbrace{(A^T A)^{-1} (A^T A)}_I x = \underline{Ax = b}$$

$$\boxed{Pb = b}$$



$$b = p + e$$

$$p = Pb$$

$$\underline{\underline{e = (I - P)b}}$$

Least squares example

(1, 1)

(2, 2)

(3, 2)

$$c + d = 1$$

$$c + 2d = 2$$

$$c + 3d = 2$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Minimize sum of squares of errors.

$$\|Ax - b\|^2 = \|e\|^2$$

$$e_1^2 + e_2^2 + e_3^2$$

$$\Rightarrow \begin{aligned} p_1 &= Pb_1 \\ p_2 &= Pb_2 \\ p_3 &= Pb_3 \end{aligned}$$

$$\text{Find } \hat{x} = \begin{bmatrix} \hat{c} \\ \hat{d} \end{bmatrix}, P$$

$$A^T A \hat{x} = A^T b \quad \leadsto \text{closest estimation}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \hat{x} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \hat{x} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \hat{\mathbf{z}} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 3 & 6 & 5 \\ 0 & 2 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{cc|c} 3 & 0 & 2 \\ 0 & 2 & 1 \end{array} \right] \quad \begin{array}{l} 3c + 6d = 5 \\ 6c + 14d = 11 \end{array}$$

$$c = 2/3 \quad d = 1/2$$

$$y = 2/3 + 1/2 t \quad \rightarrow \text{best fit line!}$$

Using calculus $\rightarrow$

$$\|e\|^2 = (c+d-1)^2 + (c+2d-2)^2 + (c+3d-2)^2$$

$$\frac{d}{dc} \|e\|^2 = 2(c+d-1) + 2(c+2d-2) + 2(c+3d-2) = 0$$

$$\Rightarrow 3c + 6d = 5$$

$$\frac{d}{dd} \|e\|^2 \Rightarrow 6c + 14d = 11$$

$$e_1 = -1/6$$

$$e_2 = 2/6$$

$$e_3 = -1/6$$

$$P + e = b$$

$$\begin{bmatrix} 7/6 \\ 10/6 \\ 13/6 \end{bmatrix} \times \begin{bmatrix} -1/6 \\ 2/6 \\ -1/6 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

↳ Add to 5

They are perpendicular.!

$$e \perp C(A)$$

If A has independent columns then $A^T A$ is invertible.

↳ If A has columns that are linearly dependent then

$A^T A \rightarrow$ has columns that are dependent hence non-invertible

$\Rightarrow$ Suppose $A^T A x = 0$

↳ must be zero

when $A^T A$ is invertible

$$\Rightarrow x^T A^T A x = 0$$

$$(Ax)^T (Ax) = 0$$

↳ $Ax = 0$

Now $A^T A x = 0$ then $Ax = 0$

and if A has independent columns then $x = 0$!!

Columns are definitely independent
if they are perpendicular unit
vectors orthonormal

eg.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$